

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. BPT-603: Elements of Nuclear Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **five** questions as given below:

[5 × 4 = 20]

- (a) All nuclides are made up of a neutron-proton mixture that we can call nuclear matter. What is its density?
- (b) Which of the following reactions are possible and why?
- (i) $n + n \rightarrow \Lambda^0 + K^+$
- (ii) $p + p \rightarrow K^+ + \Sigma^+$
- (c) How much energy is necessary to split up an alpha particle into its constituent nucleons? (Given: $M_n = 1.008665$ u, $M_p = 1.007825$ u, $M(^4\text{He}) = 4.00260$ u).
- (d) Discuss the Q -value formula for all three kinds of Beta decay.
- (e) Carbon-11 transforms by β^+ decay. Determine the products of the reaction and the energy released. Ignore the mass of the emitted neutrino. (Given: mass of $^{11}\text{C} = 11.011434$ u, mass of $^{11}\text{B} = 11.009305$ u).
- (f) A relic is found to give an activity count of 12 c.p.m. (counts per minute) for each gram of carbon. If living trees give a count of 16 c.p.m., find the approximate age of the relic. (Take the half-life of Carbon-14 as 5730 years).

Question 2.

- (a) Discuss experimental methods for the charge radius measurement of nuclei. Explain with a proper diagram. Also estimate the minimum energy of the incident ion for determining the radius of the target nucleus. [7.5]
- (b) Explain the need for introducing the colour degree of freedom for the quarks. [5]

Question 3.

- (a) Discuss the different properties of the nuclear force in detail along with supporting experimental evidence. [7]
- (b) The decay of ^{253}Es ($I = 7/2$, $\pi = +$) leads to a sequence of negative-parity states in ^{249}Bk with $I = 3/2, 5/2, 7/2, 9/2, 11/2, 13/2$. For each state, find the permitted values of the orbital angular momentum l of the emitted alpha particle. [5.5]

OR

- (a) Explain the working of the Van de Graaff (VdG) accelerator using a suitable diagram. [7.5]
- (b) Describe the expression for the Coulomb potential barrier in Alpha decay. Calculate the Coulomb potential barrier experienced by an Alpha particle in the Alpha decay of ^{226}Ra ($Z = 88$). [5]

Question 4.

- (a) An even- Z , even- N nucleus has the following sequence of levels above its 0^+ ground state: 2^+ (89 keV), 4^+ (288 keV), 6^+ (585 keV), 0^+ (1050 keV), 2^+ (1129 keV). Draw an energy level diagram and show all probable γ transitions and their dominant multipole assignments. [6.5]
- (b) For the following γ -transitions, give all permitted multipoles and indicate which multipole might be the most intense in the emitted radiation:
- (i) $9/2^+ \rightarrow 7/2^+$
 - (ii) $1/2^- \rightarrow 7/2^-$
 - (iii) $1^- \rightarrow 2^+$
 - (iv) $4^+ \rightarrow 2^+$
 - (v) $11/2^- \rightarrow 3/2^+$
 - (vi) $3^+ \rightarrow 3^+$

[6]

OR

- (a) A beam of 8 MeV α -particles is incident upon a thin gold screen ($Z = 79$). Calculate the impact parameter for scattering an incident α -particle through an angle of 110° . [4.5]
- (b) Give the quark constituents of the members of the meson 0^- octet, baryon $1/2^+$ octet, and baryon $3/2^+$ decuplet families. [8]

Question 5.

- (a) Discuss the photomultiplier tube (PMT) using a suitable diagram. Also explain gain and quantum efficiency. [4.5]
- (b) In the semi-empirical mass formula, what experimental evidence does the pairing term δ account for? For a given value of the mass number A , nuclear masses can be expressed as a quadratic function in Z . Using the minimum isobar formula, determine whether ^{142}Xe ($Z = 54$) is β^+ or β^- unstable. [8]

[Click here to visit our website](#)

[Click here to visit our contributor](#)

[Click here to visit our contributor](#)